type of cells (1.5V) are used in clocks and remote controls. 1.5 V, 3 V button cells are used in watches and computers, respectively. 9 V cell is used in multimeter and cordless microphone.

Secondary Cells

The cells that can be recharged are called as 'secondary cells'. Some of the secondary cells are lead-acid cell, Lithium ion cell and Nickel Cadmium cell.

Figure 2.2(b) shows the structure of secondary cell. Lead-acid cell is widely used in two-wheel and four-wheel vehicles. Lithium ion cells are used in laptops and cell phones. Nickel cadmium cells are used in flash lights, emergency lights and toys.

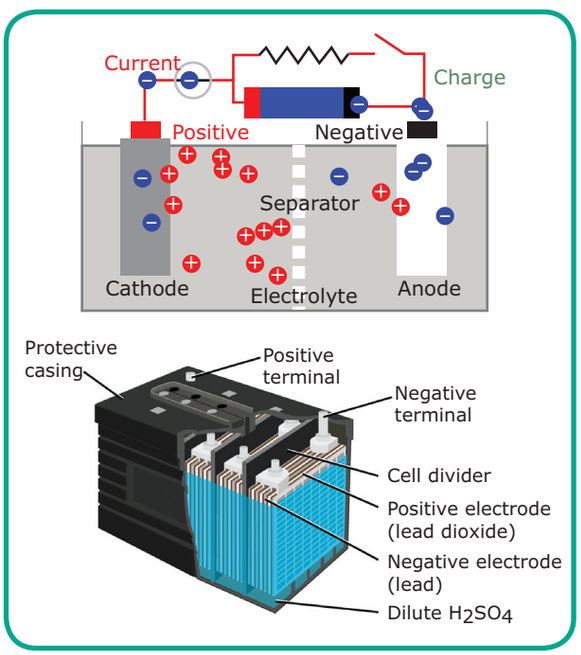


Figure 2.2(b) Structure of Secondary Cell

2.1.2. Capacity of Cells

The capacity of the cells is termed in Ampere-hour (Ah). This denotes the supply of current based on time.

For example, if a cell has a capacity of 2 Ah, it could supply the current of one Ampere in 2 hours or 2 Amperes in 1 hour.

2.1.3 Cell Arrangements

Cell arrangement determines the voltage and current. For that, the cells are connected in series and parallel.

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2.1.3.1 Cells in Series

In this method, the positive (+) terminal of one cell is connected with the negative (-) terminal of next cell and so on.

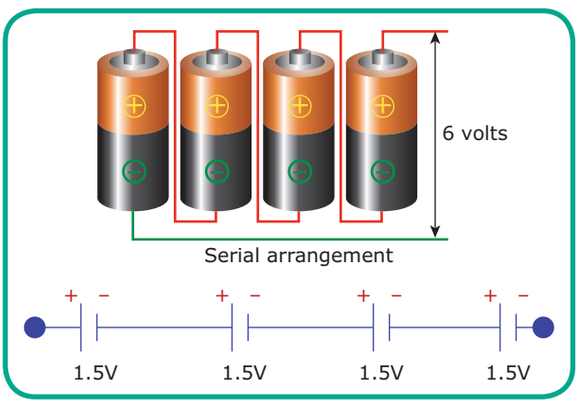


Figure 2.3 Cells Connected in Series

Figure 2.3 shows the series connection of cells. Cells have a particular potential difference (Eg.1.2 V, 1.5 V, etc.). If higher voltage is needed (E.g. 6 V, 9 V, 12 V) cells are connected in series. The current flow through in each cell is equal and same.

If four cells are connected in series (1.5 V each) the total EMF is,

$$E_T = n \times e$$
$$= 4 \times 1.5 = 6 \text{ V}$$

where '*n*' is number of cells and '*e*' is the voltage of the cell.

2.1.3.2 Cells in Parallel

In this method, the positive terminal of each cell is connected simultaneously and

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